

SEA & BEYOND

The career platform the maritime industry deserves

Technology Partnership Requirements Document

Confidential

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The Vision

Sea & Beyond is a career platform for the global maritime community. Not a job board, it's a lifelong career co-pilot. We are building a single platform where seafarers can find the right jobs, upskill, get mentored, connect with peers, and access maritime products and services.

The closest parallel: combine LinkedIn's professional networking, Coursera's structured learning, and the kind of personalised career guidance that simply does not exist in maritime today. The platform serves seafarers at every stage - cadets entering the industry, mid-career officers, and senior professionals exploring shore-based transitions.

We have deep domain expertise, a clear product roadmap, and a UI/UX team already working on the design system and user flows. What we need is a technology partner to build and scale this with us (a long-term collaboration, not a one-off project.)

How We Are Approaching the Build

We are building in phases. Each phase delivers a working product increment and carries its own scope, including any AI capabilities relevant to that phase.

Phase 1: Sea-Going Vertical

Candidate Interface + Recruiter Interface + Admin Panel. Together these form a functional marketplace for sea-going jobs. This is our immediate priority and the first product we take to market. This includes the AI features needed to make the sea-going vertical intelligent from day one; document parsing, smart job matching, and candidate recommendations.

Phase 2: Community Platform

Forums, Q&A, knowledge sharing, and peer networking. The Community serves the entire Sea & Beyond ecosystem, not just the sea-going vertical. It drives organic engagement and gives seafarers a reason to stay active on the platform beyond job hunting. AI capabilities relevant to the community - content recommendations, intelligent moderation, expert matching - are part of this phase.

Future Phases

LMS, mentorship, commerce, communication layer, shore jobs vertical, and advanced analytics. Phasing for these will be discussed later, but this document covers their scope so that architecture decisions made now account for everything ahead.

A note on AI: AI is not a separate phase - it is woven into every phase. Each phase carries its own AI requirements, and those get built alongside the core features of that phase. The architecture should support this from the start: clean data pipelines, structured event logging, modular ML service boundaries, and APIs designed for intelligence to plug into naturally.

A note on deliverables: Each phase is expected to deliver both a web platform and mobile applications (Android and iOS). Our users are primarily on mobile, and the app is not an afterthought - it is a core deliverable alongside the web experience.

A note on design: UI/UX designs, user flows, wireframes, and design specifications will be provided by our design team. The technology partner implements these designs and advises on technical feasibility where needed.

Platform Modules

Below is the complete set of modules that make up Sea & Beyond, along with the phase they belong to.

3.1 Candidate Interface [Phase 1]

The primary interface for seafarers. This is where they manage their careers on the platform.

Onboarding and Profile

- User registration via social login (LinkedIn, Google) and email/phone
- Profile creation with structured fields for maritime credentials, sailing experience, certifications, and documents
- Document upload and secure storage (CDC, COC, certificates, resume)
- AI-powered document parsing to auto-extract and populate profile fields from uploaded certificates and resumes
- Profile completeness indicator and readiness checks

Job Discovery and Application

- Job search and filtering by rank, vessel type, company, location, contract duration, and other relevant parameters
- AI-powered job matching that considers rank, experience, vessel type, certifications, availability, location, and company fit
- Match scoring with explanations for recommended jobs
- Job saves, alerts, and notifications for new relevant openings
- Application submission and full lifecycle tracking (Applied, Shortlisted, Interview, Offer, Joined)
- Application history and status visibility

Dashboard and Career Tools

- Personalised dashboard surfacing pending actions, application updates, new matches, and relevant activity
- Role and company comparison tools
- Salary benchmarks and market insights
- Career timeline showing sailing history and progression

3.2 Recruiter Interface [Phase 1]

The interface for shipping companies and manning agents to manage hiring.

Job Management

- Create, edit, duplicate, and close job listings
- Job templates for common maritime roles
- Job visibility controls (public, invite-only, internal)
- Bulk operations for managing multiple openings

Candidate Discovery and Management

- Candidate search with filters (rank, vessel type, certifications, experience, availability, location)
- AI-assisted candidate recommendations for open positions
- Candidate profile viewing with key details
- Shortlisting, bookmarking, and internal notes
- Application pipeline management with stage tracking
- In-platform communication with candidates
- Interview scheduling

Company Presence and Analytics

- Company profile pages (about, fleet, culture, open positions)
- Review and rating system from seafarers
- Hiring funnel metrics and analytics (views, applications, shortlists, offers, joins)
- Time-to-hire and source effectiveness tracking

3.3 Admin Interface [Phase 1]

Internal administration panel for the Sea & Beyond team to manage and operate the platform.

User and Account Management

- User account management (view, search, edit, suspend, delete) for candidates and recruiters
- Recruiter and company verification and onboarding approval workflow
- Role-based access control (super admin, operations, content moderator, support)
- Admin action audit trail

Content and Listing Oversight

- Job listing review and approval workflow
- Content moderation queue for community posts, reviews, and comments (applicable when community goes live in Phase 2)
- Ability to flag, hide, or remove content with reason tracking

- Master data management: vessel types, rank hierarchies, certification databases, job categories

Analytics and Reporting

- Platform health dashboard: user metrics, job activity, application volume
- User engagement metrics (signups, active users, retention, feature adoption)
- Recruiter performance metrics
- Conversion funnel tracking
- AI model performance monitoring: match accuracy, parsing quality, recommendation effectiveness
- Exportable reports (CSV, PDF)

System Configuration

- Feature toggles
- Notification and email template management
- System health monitoring and error logs

3.4 AI Capabilities [Phase 1]

The following AI/ML capabilities are required as part of Phase 1 and should be built alongside the candidate, recruiter, and admin interfaces.

- Document intelligence: parsing CDC, COC, certificates, and resumes using NLP/NER to auto-extract structured data and populate candidate profiles
- Smart job matching: ML-based matching engine that scores and ranks jobs for candidates (and candidates for jobs) based on rank, experience, vessel type, certifications, availability, location, and company fit
- Match explanations: generating human-readable reasons for why a job or candidate was recommended
- Candidate recommendations for recruiters: surfacing the best-fit candidates for open positions
- Data pipeline foundation: structured event logging, user behaviour tracking, and clean data models that feed the ML layer and support future AI enhancements across all phases

3.5 Community Platform [Phase 2]

The engagement layer that gives seafarers a reason to return beyond active job searching. The Community serves the entire Sea & Beyond ecosystem.

- Stage-based and interest-based communities (cadets, officers, shore professionals, topic groups)
- Q&A forums with threading, upvoting, and expert recognition
- Knowledge sharing: articles, experience posts, career tips from members

- Events and webinars: registration, reminders, recordings
- Peer networking and direct messaging
- Achievement and milestone celebrations
- Public view-only access for non-logged-in users; full participation requires an account

3.6 AI Capabilities [Phase 2]

AI features relevant to the community platform, built alongside Phase 2.

- Content recommendations: surfacing relevant posts, discussions, and articles based on a member's profile, interests, and activity
- Intelligent moderation: automated flagging of inappropriate or spam content for moderator review
- Expert matching: identifying and highlighting knowledgeable contributors within communities
- Engagement signals: using behavioural data to personalise community feeds and notifications

3.7 AI Career Agent [Future Phase]

The advanced intelligence layer that evolves Sea & Beyond from a platform into a personalised career co-pilot.

- Deep profile and behavioural analysis to build comprehensive understanding of each user
- Personalised career roadmap generation with milestones and skill gap identification
- Continuous progress tracking and behavioural learning
- Contextual next-best-action recommendations (jobs, courses, mentors, connections)
- Predictive career path modelling

3.8 Learning Management System [Future Phase]

Structured learning integrated with career progression.

- Course catalogue and content management
- Video hosting and progress tracking
- Assessment and quiz engine
- Certificate generation and issuance
- AI-driven course recommendations tied to identified skill gaps

3.9 Mentorship Module [Future Phase]

Connecting experienced maritime professionals with those earlier in their careers.

- Mentor-mentee matching based on career stage, goals, and expertise
- Session booking and scheduling
- Video call integration
- Session history, feedback, and rating system

3.10 Commerce [Future Phase]

A marketplace layer for maritime products, services, and industry intelligence.

- Curated marketplace for maritime products and services
- Company directory with detailed profiles
- Salary data, market trends, and industry insights

3.11 Communication Layer [Future Phase]

Multi-channel communication to keep users engaged and informed.

- WhatsApp Business API integration for alerts, updates, and two-way messaging
- Push notifications for job alerts, application updates, and community activity
- Email system: transactional emails, newsletters, and digests
- All channels asynchronous and free for users

3.12 Shore Jobs Vertical [Future Phase]

Phase 1 focuses on sea-going jobs. The Shore Jobs Vertical extends the platform to cover shore-based maritime roles, serving seafarers looking to transition ashore as well as professionals already in shore-side positions.

Shore roles span a wide range — port operations, ship management, maritime law, classification societies, logistics, maritime education, and more. The candidate profiles, job matching parameters, and recruiter needs differ from the sea-going side.

- Shore-specific candidate profiles with relevant fields (shore experience, industry sub-sector, functional area, qualifications)
- Shore job categories and role taxonomy distinct from sea-going roles
- Adapted AI matching logic accounting for shore-specific parameters
- Recruiter tools tailored for shore hiring workflows
- Career transition support for sea-going professionals moving into shore roles

- Integration with the AI Career Agent for shore-specific roadmaps and recommendations
- Shared community and networking layer across both verticals

3.13 Advanced Analytics and Optimisation [Future Phase]

- AI model performance monitoring and drift detection across all modules
- Predictive analytics for user behaviour and career outcomes
- A/B testing infrastructure
- Advanced recruiter and platform-level reporting

Technical Considerations

Key context about our users and what it means for technical decisions:

Mobile-first, with native apps. The majority of our users are on mobile. The deliverable for each phase includes both a web platform and mobile applications (Android and iOS). The mobile app is a core product, not a wrapper or an afterthought.

Offline and low bandwidth. Seafarers deal with slow, expensive, and intermittent satellite connectivity. PWA capability, offline access for critical features, compressed assets, lazy loading, and minimal data transfer are essential.

AI built into every phase. Each phase carries its own AI/ML requirements. The architecture needs to support ML services from the start — clean data pipelines, structured event logging, modular ML service boundaries, and APIs designed for intelligence to plug into naturally. This is not a bolt-on; it is part of the core platform.

Scalable and modular. The platform grows module by module. The architecture should allow independent scaling and deployment of services.

Cross-platform mobile framework. Android and iOS from a shared codebase. We are open to recommendations on framework (React Native, Flutter, or otherwise).

Tech stack is open. No constraints on cloud provider, language, or framework. We want your recommendation based on what fits this product best.

UI/UX provided separately. Our design team delivers wireframes, flows, and specs for web and mobile. The development partner implements and advises on feasibility.

What We Need From You

Review this document and respond with the following:

1. Technical Approach

Your recommended tech stack for web, mobile, backend, AI/ML, and infrastructure. Architecture approach and how you would structure the system to be modular, scalable, and AI-integrated from Phase 1. Your take on the admin panel. How you would handle offline capability and low-bandwidth optimisation. Your approach to the AI/ML components - in-house models, third-party APIs, or a hybrid.

2. Team Structure

Who would work on this, what roles, and how many people. We expect the team to cover web, mobile, backend, AI/ML, and QA. Relevant experience with marketplace platforms, community products, AI-integrated applications, or mobile-first builds. Specific project examples where possible.

3. Build Plan

How you would approach building Phase 1 and Phase 2, followed by subsequent phases. Milestones, deliverables (web + mobile), and a realistic timeline for each phase. How AI capabilities would be developed alongside core features in each phase.

4. Timeline

When can Phase 1 reach a launchable state (web + mobile)? How long after that for Phase 2? Estimated timeline for the full platform.

5. Questions and Recommendations

Clarifications you need from us. Risks or challenges you foresee. Recommendations based on your experience building similar products.

We are looking for a technology partner, not a vendor. If this is the kind of challenge you enjoy, we would love to hear from you.

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